

Life Cycle Policies

1. Create a new S3 bucket named 'my-zomato-images' and upload at least 3 sample image files representing restaurant menu items.

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1. Log in to the AWS Management Console.
 2. Open Amazon S3.
 3. Click Create Bucket.
 4. Enter the bucket name:
 - my-zomato-images

➤ Note: S3 bucket names must be globally unique. If this name is already in use, create a unique variation such as:
 5. Select the Asia Pacific (Mumbai) region.
 6. Click Create Bucket.
 7. Open the bucket and click Upload.
 8. Upload at least three sample restaurant menu images, for example:
 - pizza.jpg
 - burger.jpg
 - pasta.jpg

➤ Result: The bucket contains three uploaded image files.
2. Set up a lifecycle policy on your 'my-zomato-images' S3 bucket to automatically transition all objects older than 30 days to the Glacier storage class.

Hint: Use the S3 Management Console's 'Lifecycle rules' section to define the transition action.

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1. Open the **my-zomato-images** bucket.
2. Select the **Management** tab.
3. Click **Create Lifecycle Rule**.
4. Enter the rule name:
 - MoveToGlacierAfter30Days
5. Apply the rule to **all objects** in the bucket.

6. Under **Lifecycle actions**, select:
 - **Transition current versions of objects**
 - Storage Class: **S3 Glacier Flexible Retrieval**
 - Days after object creation: **30**
 7. Save the lifecycle rule.
- **Result:** Any object older than **30 days** is automatically moved to the Glacier storage class.
3. Simulate a cost optimization scenario: calculate and compare the estimated monthly storage cost for keeping 5GB of images in S3 Standard versus S3 Glacier using AWS Pricing Calculator.
Hint: Use the AWS Pricing Calculator website and note down the difference in INR.

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- Using the AWS Pricing Calculator, the estimated monthly storage cost for **5 GB** is approximately:

Storage Class	Estimated Monthly Cost (INR)*
Amazon S3 Standard	₹10–₹15
Amazon S3 Glacier	₹2–₹5

- Prices vary by region and current AWS pricing.
- **Conclusion:** S3 Glacier is significantly less expensive than S3 Standard for long-term storage, making it a better choice for infrequently accessed files.
4. Add a new rule to your S3 bucket's lifecycle policy so that any objects with the prefix 'offers/' (e.g., 'offers/summer2024.jpg') are deleted automatically after 60 days.

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1. Open Lifecycle Rules.
2. Edit the existing rule or create a new rule.
3. Apply the rule only to objects with the following prefix:
4. Under Expiration, choose:
 - Delete current versions of objects
 - 60 days after object creation

5. Save the rule.

- Result: Any object stored inside the offers/ folder (for example, offers/summer2024.jpg) is automatically deleted after 60 days.

5. Use ChatGPT to generate a sample JSON configuration for an S3 lifecycle policy that transitions files to Glacier after 15 days and deletes them after 180 days. Paste the generated JSON and briefly explain each section.

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- **Explanation**

Section	Description
ID	The name of the lifecycle rule.
Status	Enables or disables the rule.
Filter	Specifies which objects the rule applies to. An empty prefix means all objects in the bucket.
Transitions	Moves objects to the Glacier storage class after 15 days .
Expiration	Permanently deletes the objects after 180 days .